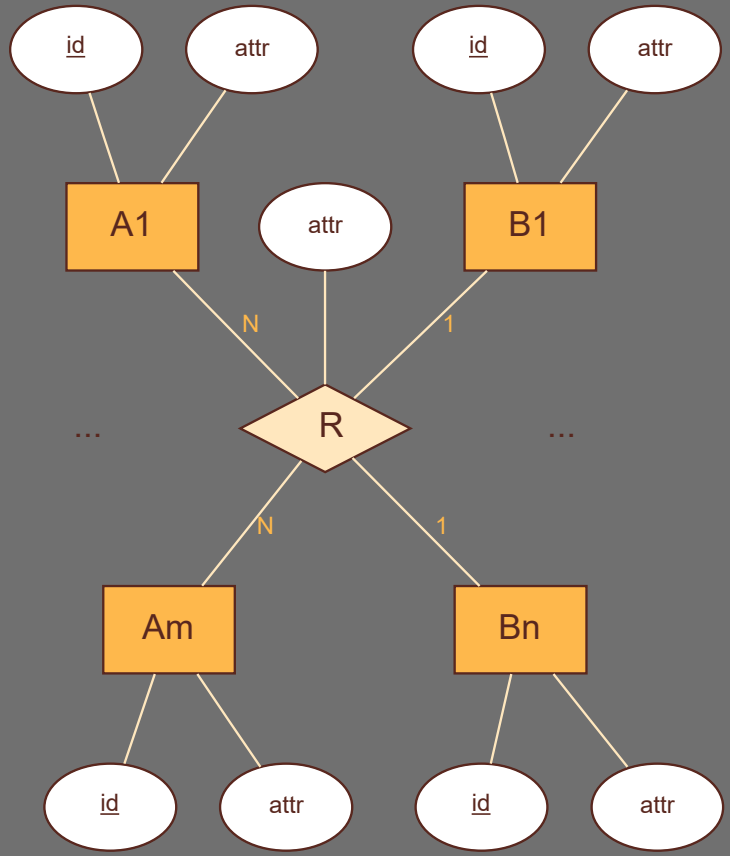
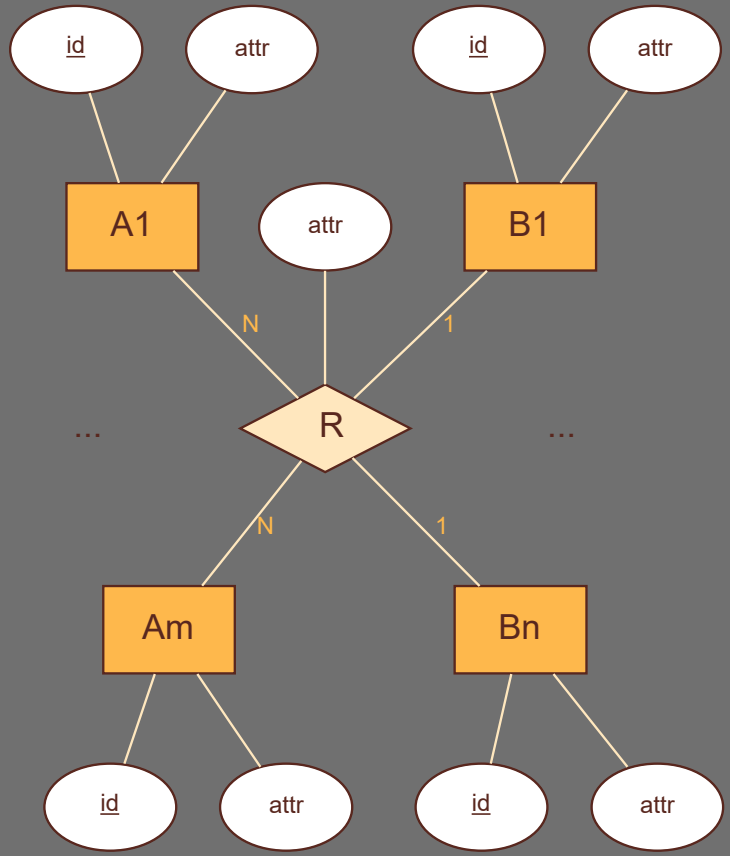
ERD Implementation Basics: Generalizing & Abstracting



General form

Solution depends on m.

- m>1
- m=1
- m=0



Case: m>1

Like many-many.

$A_1(\underline{id}, attr)$

...

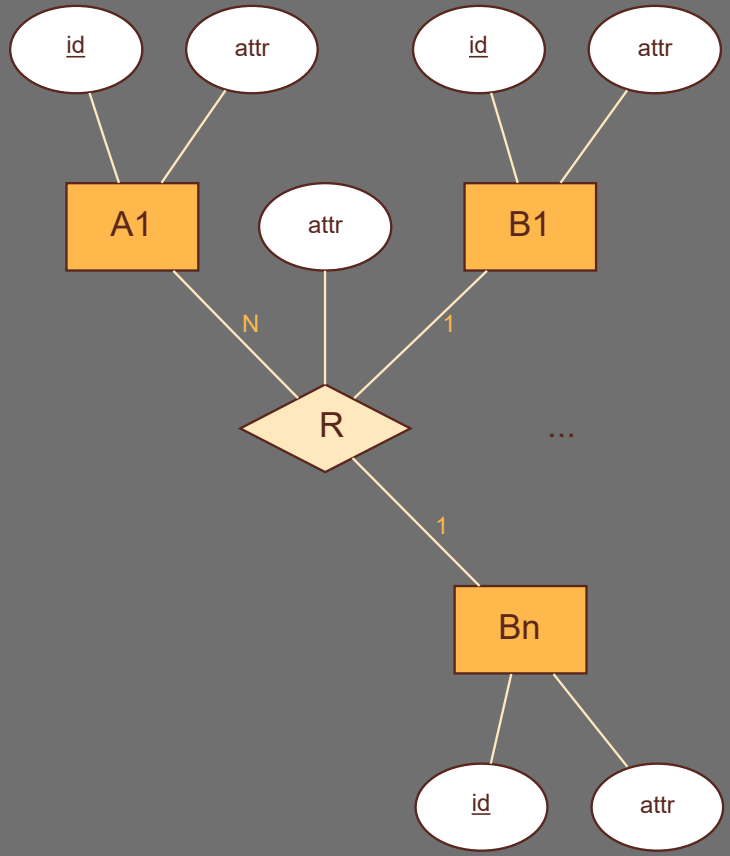
$A_m(\underline{id}, attr)$

$B_1(\underline{id}, attr)$

...

$B_n(\underline{id}, attr)$

$R(\underline{a_1_id}, ..., \underline{a_m_id}, b_1_id, ..., b_n_id, attr)$



Case: m=1

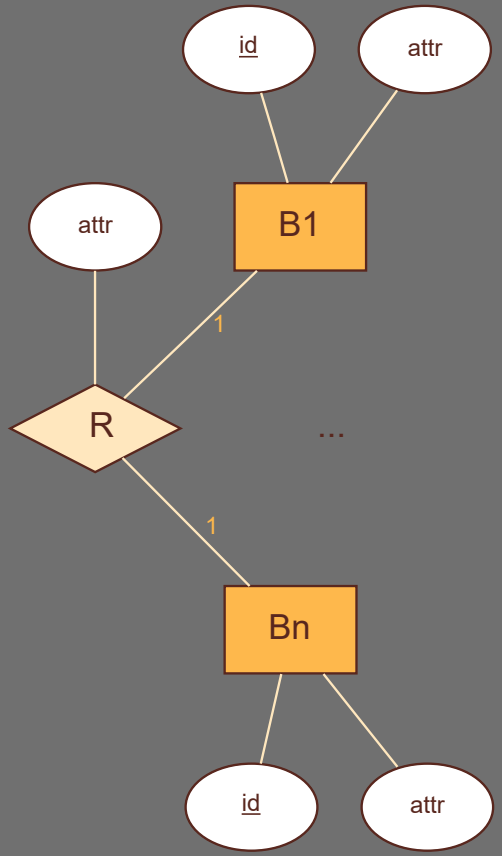
Like 1-many.

$A_1(\underline{id}, attr, b_1_id, \ldots, b_n_id, r_attr)$

$B_1(\underline{id}, attr)$

...

$B_n(\underline{id}, attr)$



Case: m=0

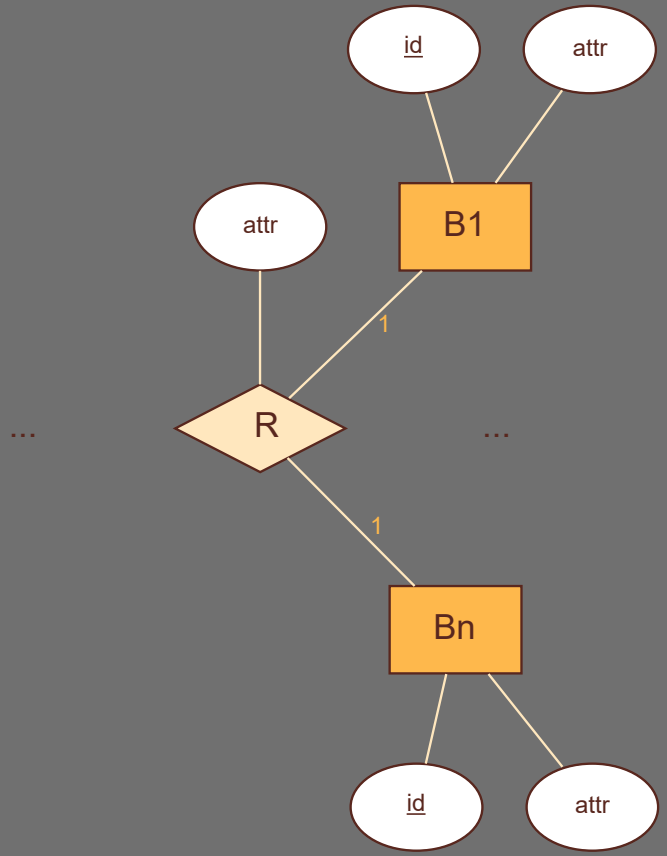
Option 1

Like 1-1.

$B_1(\underline{id}, attr, b_2_id, \ldots, b_n_id, r_attr)$

...

$B_n(\underline{id}, attr, b_1_id, \ldots, b_{n-1}_id)$



Case: m=0

Option 2

Like many-many.

$B_1(\underline{id}, attr)$

...

$B_n(\underline{id}, attr)$

$R(\underline{b_1_id}, \ldots, \underline{b_n_id}, attr)$

Summary

General form looks complicated.

However,

- Rarely need beyond ternary.
- 1-1 is uncommon, 1-1-1 is rare.